

Thyroid hormone replacement therapy

If you have had surgery to remove your thyroid gland, you will need to take hormone replacement therapy.

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What is thyroid hormone replacement therapy?

[Surgery](#) can be used to treat [papillary and follicular thyroid cancers](#), [medullary thyroid cancer \(MTC\)](#) and sometimes [anaplastic thyroid cancer](#).

If you have all of your [thyroid gland](#) removed, you will need to take thyroid hormone replacement tablets every day for the rest of your life. This replaces the thyroxine (also known as T4) which your thyroid gland is not making.

Thyroid hormone tablets do 2 very important things. They help to:

- keep your body functioning at the correct speed (metabolism)
- reduce the risk of thyroid cancer coming back.

If you had a low risk thyroid cancer and only had part of the thyroid removed you might not need replacement hormones.

But even if only part of the thyroid gland is removed, some people may need to take thyroid hormone replacement tablets. Sometimes you may not need to take them immediately. Instead, you will start to take them after a blood test.

Taking thyroid hormone tablets

You usually have a hormone drug called levothyroxine. It is a synthetic version of the thyroxine (T4) your thyroid gland is not making. It replaces T4 after your [thyroid](#) is removed. Levothyroxine has different brand names.

Hormone replacement tablets help to prevent the signs and symptoms of hypothyroidism. These include:

- tiredness
- weight gain
- feeling slowed down both physically and mentally
- feeling cold
- dry skin and hair
- hair thinning
- constipation.

Finding the correct dose

The dose you start taking depends on your weight. Your doctor might change this depending on the results of regular blood tests that measure your T4 and TSH levels.

You will have regular blood tests to check the levels of thyroid hormones. Your cancer team will monitor these to make sure you are having the correct dose.

It might take a few months to find the right dose of thyroid hormones for you. During this time, you may have some symptoms such as [tiredness](#).

Once the correct dose is found, you should not have any side effects from the tablets. This is because they are replacing the hormones your thyroid gland would have made naturally.

How to take thyroid hormone tablets

Your doctor, nurse or pharmacist will talk to you about when to start taking your thyroid hormone replacement tablets.

Always take them exactly as they explain. This helps make sure they work as well as possible for you.

It is important to check with your pharmacist that the type of thyroid hormone tablets they give you are exactly as prescribed. This is to try and make sure you continue to take the same brand.

If you have any questions about your medication, speak to your doctor, specialist nurse or pharmacist.

Other things to remember about taking your thyroid hormone tablets:

- Take the tablets at the same time every day
- Take them at least 30 minutes before breakfast or your first meal of the day without any other medications.
- Do not take levothyroxine together with calcium and zinc – wait for at least 4 hours before taking these supplements. This is because they might affect the way your body absorbs the thyroid hormone replacement tablets.
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Swallow the tablets with plenty of water.

Pregnancy

If in the future you are planning on getting pregnant, talk to your thyroid doctor (endocrinologist) or your cancer doctor first. They can explain:

- changes to your levothyroxine dose in pregnancy
- how often your thyroid hormone levels will be checked.

This helps look after your health, and the wellbeing of the baby.

Reducing the risk of thyroid cancer coming back

Thyroid hormone drugs like levothyroxine stop your body making thyroid-stimulating hormone (TSH). TSH helps the healthy thyroid to produce thyroid hormones. But if you had [papillary or follicular thyroid cancer](#) it can encourage thyroid cancer cells to grow.

After surgery you will be given levothyroxine at a slightly higher dose to keep TSH levels very low. Doing this helps to reduce the risk of the cancer coming back after surgery. This is called TSH suppression and is part of your ongoing treatment.

It is most important in the first 2 years. After this time and if you are in [a low risk group](#) your cancer doctor may talk to you about allowing TSH levels to increase a little.

About our information

References



Below is a sample of the sources used in our thyroid cancer information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

British Medical Journal. Best Practice Guidelines, Thyroid cancer. 2020.

European Society Medical Oncology (ESMO): Thyroid cancer, Clinical Practice Guidelines for Diagnosis, Treatment and Follow-up. 2019.

National Institute for Health and Care Excellence (NICE). TA535: Lenvatinib and Sorafenib for treating differentiated thyroid cancer after radioactive iodine. 2018. www.nice.org.uk/guidance/ta535 [accessed May 2021].

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, [Professor Nick Reed](#), Consultant Clinical Oncologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

An anonymous network of people affected by thyroid cancer which is free to join. Share experiences, ask questions and talk to people who understand.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

[Get help](#)